

MA-RAG: Multi-Domain Hallucination-Aware Agentic RAG Framework

Classification & Routing Layer

User Query
(Text Case / Question)

Medical Pipeline (Multi-Round Agentic RAG Debate)

Stage 1: Multi-Agent Parallel Generation
(Expert A: Primary Care, Expert B: Specialist, Expert C: Emergency)

Stage 2: Peer-Review & Debate Critique
(Cross-critique for inaccuracies & hallucinations)

External KB Retrieval
(BM25 + Rerank)

Stage 3: Moderator Synthesis & Consensus Editing
(Consolidates final stances into a clean draft)

Disagreement Query Generator

Consensus Reached & High/Med Confidence?

Approved Medical Response

Master Domain Classifier
(Router Agent)

Safety Block
(Immediate Reject)

Legal Agent
(Statutes & Precedents RAG)

Financial Agent
(Risk & Portfolio RAG)

Master Summarizer Agent
(Multi-Domain Response Synthesis)

Consolidated Final Expert Response
(Factually Verified & Hallucination Checked)

Legend:

- Medical debate steps
- Final response output
- Safety block
- Routing & Core Agents
- Other Domain agents